EQUIPFLOW QA ASSESSMENT

Before & After QA Test Case Report & Database Migration Summary

Generated: 2026-08-13 | Target DB: MongoDB Atlas

1. Executive Summary

EquipFlow is an enterprise-grade Asset & Equipment Reservation Platform. This document captures the complete verification lifecycle, including the seamless migration from SQLite to MongoDB Atlas, the 19-point Vitest API Automation Test Suite execution results, the deliberate Red Run failure capture, and the AI Change-Loop log.

2. Architecture & Database Migration (Before vs After)

Metric / Component	BEFORE (SQLite Local)	AFTER (MongoDB Atlas Cloud)
Database Engine	better-sqlite3 (Local File)	**MongoDB Atlas (Cloud Cluster)**
Connection Driver	Synchronous SQLite WAL	**Mongoose 8 async connection**
Primary Keys	Auto-increment Integer (1, 2, 3)	**ObjectId ("60c72b2f...")**
Environment Config	data/equipflow.db local file	**mongodb+srv://... in .env**
Test Execution	19/19 Passed (279ms)	**19/19 Passed (15.78s)**
Data Scalability	Single-node filesystem bound	**Cloud Distributed Cluster**

3. Test Suite Verification Results (19/19 PASS)

Category	Test Scenario & Expectation	Status
1. Happy Path	GET /api/assets -> Seeded inventory list returned	**PASS [HTTP 200]**
1. Happy Path	GET /api/health -> Health status check	**PASS [HTTP 200]**
1. Happy Path	VIP User auto-confirmed reservation (No Admin queue)	**PASS [HTTP 201]**
1. Happy Path	Standard User reservation -> Enters PENDING queue	**PASS [HTTP 201]**
1. Happy Path	Admin can approve PENDING reservation	**PASS [HTTP 200]**
1. Happy Path	Admin can reject PENDING reservation	**PASS [HTTP 200]**
1. Happy Path	User can cancel own PENDING reservation	**PASS [HTTP 200]**
2. Business Rules	CONFLICT: Overlapping dates rejected	**PASS [HTTP 409]**
2. Business Rules	QUOTA: Standard user blocked at >3 active bookings	**PASS [HTTP 400]**
2. Business Rules	BLACKOUT: Booking blocked during admin maintenance window	**PASS [HTTP 409]**
2. Business Rules	PENALTY (VIP): 3 days overdue -> $45 (50% VIP discount)	**PASS [HTTP 200]**
2. Business Rules	PENALTY (Standard): On-time return = $0 penalty	**PASS [HTTP 200]**
3. Validation	REJECT: Inverted dates (start > end)	**PASS [HTTP 409]**
3. Validation	REJECT: Booking duration > 14 days limit	**PASS [HTTP 409]**
3. Validation	REJECT: Non-existent asset ID returns 404	**PASS [HTTP 404]**
3. Validation	REJECT: Unauthenticated request returns 401	**PASS [HTTP 401]**
3. Validation	REJECT: Standard user cannot create blackouts	**PASS [HTTP 403]**
3. Validation	REJECT: Missing required fields (Zod 400)	**PASS [HTTP 400]**
3. Validation	REJECT: Invalid date format (Zod 400)	**PASS [HTTP 400]**

4. Deliberate Red Run Verification

As required by Stage 2 of the Tactive Assessment, a regression was deliberately injected into conflictEngine.ts to verify that the automated test suite catches regressions. The test suite correctly flagged the double-booking flaw with an assertion failure (expected HTTP 409, got HTTP 201).

ðŸ”´ RED RUN EVIDENCE OUTPUT

FAIL tests/api/reservation.test.ts
x CONFLICT: Overlapping dates for same asset must be rejected (409)
AssertionError: expected 201 to be 409
Result: 1 failed, 18 passed (19 total)

5. AI Change Loop Summary

Mid-development, the VIP Priority Tier feature was added (auto-approval + 50% late fee discount). The AI Change Loop caught stale test expectations, self-corrected the test assertions, and restored 100% test pass rate in 3 iterations.

GitHub Repository: https://github.com/kathir022005/Tactive

Submission Status: COMPLETE & READY FOR EVALUATION